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FIT3155: Advanced Algorithms and Data Structures

Week 1: **Gusfield's Z-Algorithm and Exact Pattern Matching**

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What is covered here?

Linear-time approaches to the exact pattern matching problem on strings

- This week
  - Gusfield's Z-algorithm
- Next week
  - Boyer-Moore's algorithm
  - Knuth-Morris-Pratt's algorithm

Source material and recommended reading

- Dan Gusfield, Algorithms on Strings, Trees and Sequences, Cambridge University Press. (Chapters 1-2).

Exact pattern matching: Introduction

The exact pattern matching problem

Given a reference text **txt**$[1 \ldots n]$ and a pattern **pat**$[1 \ldots m]$, find **ALL** occurrences, if any, of **pat** in **txt**.

Example

If, **txt**$[1 \ldots 12] =$ b b a b a x a b a b a y (positions 1 2 3 4 5 6 7 8 9 10 11 12) and **pat**$[1 \ldots 3] =$ a b a (positions 1 2 3) then **pat** occurs in **txt** at positions 3,7 and 9.

	1	2	3	4	5	6	7	8	9	10	11	12
txt:	b	b	a	b	a	x	a	b	a	b	a	y
pat:			✓ a 1	✓ b 2	✓ a 3							
pat:							✓ a 1	✓ b 2	✓ a 3			
pat:									✓ a 1	✓ b 2	✓ a 3	

Exact pattern matching: Introduction

- Problem arises in innumerable applications
  - Word processing – `grep` command in Unix
  - Search Engines – Google
  - Library catalogs
  - ⋮

Naïve algorithm

```
for i from 1 to n-m+1:  // for each position i in txt
    // check if txt[i..i+m-1]==pat[1..m]
    for j from 1 to m:
         if txt[i+j-1] != pat[j] then
             break
         endif
    endfor
    //if pat matched fully at i, print position in txt
    if (j == m+1) print i
endfor
```

Naïve algorithm

	1	2	3	4	5	6	7	8	9	10
txt:	a	a	a	a	a	a	a	a	a	a
iteration-1	✓	✓	✓							
pat:	a	a	a							
	1	2	3							
iteration-2		✓	✓	✓						
pat:		a	a	a						
		1	2	3						
iteration-3			✓	✓	✓					
pat:			a	a	a					
			1	2	3					
⋮					⋮					
iteration-8								✓	✓	✓
pat:								a	a	a
								1	2	3

How many comparison of symbols does this approach perform in worst case?

Total number of comparisons, in the **worst case**, is $(n - m + 1) \times m = \mathcal{O}(mn)$.

In general, naïve approach involves too many unnecessary comparisons

	1	2	3	4	5	6	7	8	9	10	11	12	13
txt:	x	a	b	x	y	a	b	x	y	a	b	x	z
iteration-1	✗												
pat:	a	b	x	y	a	b	x	z					
	1	2	3	4	5	6	7	8					
iteration-2		✓	✓	✓	✓	✓	✓	✓	✗				
pat:		a	b	x	y	a	b	x	z				
		1	2	3	4	5	6	7	8				
iteration-3			✗										
pat:			a	b	x	y	a	b	x	z			
			1	2	3	4	5	6	7	8			
iteration-4				✗									
pat:				a	b	x	y	a	b	x	z		
				1	2	3	4	5	6	7	8		
iteration-5					✗								
pat:					a	b	x	y	a	b	x	z	
					1	2	3	4	5	6	7	8	
iteration-6						✓	✓	✓	✓	✓	✓	✓	✓
pat:						a	b	x	y	a	b	x	z
						1	2	3	4	5	6	7	8

Overall 20 comparisons in the naïve approach, on this example.

Ideas for speeding up the naïve method

Shift more purposefully: Shift **pat** by > 1 position after each iteration, whenever possible.

Shift always safely: Never shift so far as to miss any exact occurrence of **pat** in **txt**.

A smarter approach can reduce unnecessary comparisons

Scenario 1:

A smarter approach can gather that, after the mismatch in iteration-2 (for example), there is enough information to shift **pat** by > 1 position character in the next iteration.

	1	2	3	4	5	6	7	8	9	10	11	12	13
txt:	x	a	b	x	y	a	b	x	y	a	b	x	z
iteration-1	✗												
pat:	a	b	x	y	a	b	x	z					
	1	2	3	4	5	6	7	8					
iteration-2		✓	✓	✓	✓	✓	✓	✓	✗				
pat:		a	b	x	y	a	b	x	z				
		1	2	3	4	5	6	7	8				
iteration-3						✓	✓	✓	✓	✓	✓	✓	✓
pat:						a	b	x	y	a	b	x	z
						1	2	3	4	5	6	7	8

How does this approach achieve this?

By the end of iteration-2, the approach has learnt that **pat**$[1 \ldots 7] =$ **txt**$[2 \ldots 8]$. It can gather that **pat**$[1] =$ 'a' does not occur until position 6 in **txt**. This is enough to conclude there can be no exact matches of **pat** starting positions 3 to 6 in **txt**, and hence can shift it by $4 > 1$ positions **safely**.

Overall, this 'smarter' algorithm saves 3 comparisons on this example.

An even smarter approach can reduce unnecessary comparisons even more

Scenario 2:

An 'even smarter' algorithm can gather more information, beyond scenario 1, and render more comparisons redundant.

	1	2	3	4	5	6	7	8	9	10	11	12	13
txt:	x	a	b	x	y	a	b	x	y	a	b	x	z
iteration-1	✗												
pat:	a	b	x	y	a	b	x	z					
	1	2	3	4	5	6	7	8					
iteration-2		✓	✓	✓	✓	✓	✓	✓	✗				
pat:		a	b	x	y	a	b	x	z				
		1	2	3	4	5	6	7	8				
iteration-3									✓	✓	✓	✓	✓
pat:						a	b	x	y	a	b	x	z
						1	2	3	4	5	6	7	8

How does this algorithm achieve this?

This algorithm preprocesses **pat** and learns that **pat**$[1...3]$ ('abx') is same as **pat**$[5...7]$. After iteration 2, it knows **pat**$[5...7]$ = **txt**$[6...8]$, allowing it to shift such that **pat**$[1...3]$ aligns under **txt**$[6...8]$ without re-comparing those characters in iteration-3, saving 3 more comparisons.

Overall, this 'even smarter' algorithm saves 6 comparisons over 'naïve'.

Take home message from these illustrated examples

- The examples shown in the previous slides illustrate the kind of ideas that allow unnecessary comparisons to be skipped/jumped over.
  - Note, we haven't yet rationalized how an algorithm can **efficiently** implement such ideas shown in previous scenarios.
- Over weeks 1 and 2, we will study 3 remarkable algorithms that can be implemented to run pattern matching in **linear time**.
  - **Gusfield's Z-Algorithm**
  - **Boyer-Moore's Algorithm**
  - **Knuth-Morris-Pratt's Algorithm**

1. Gusfield's Z-Algorithm

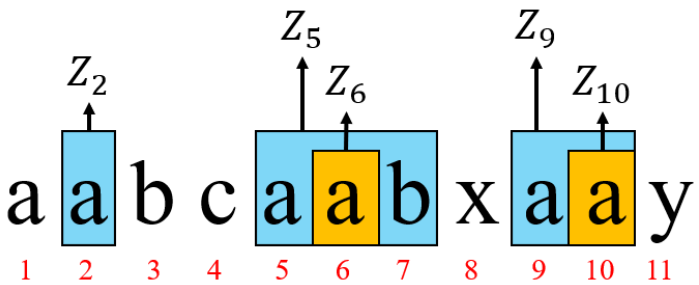
Comments:

- This is actually a linear-time algorithm to **preprocess** a given string.
- After preprocessing, the output from this algorithm (Z_i-values) can be used to address a **versatile set of problems** that arise in strings, including **linear-time pattern matching**.
- In this unit you will witness its various uses.

Gusfield's Z-algorithm – Defining Z_i

Definition of Z_i:

For a string **str**$[1 \ldots n]$, define Z_i (for each position $i > 1$ in **str**) as the **length** of the **longest substring** starting at **position** i of **str** that **matches its prefix** (i.e., **str**$[i \ldots i{+}Z_i\text{-}1] =$ **str**$[1 \ldots Z_i]$).



Example

```
      1 2 3 4 5 6 7 8 9 10 11
str= a a b c a a b x a a  y
```

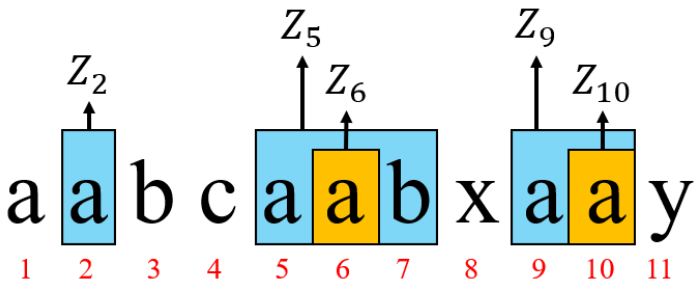
$Z_1 =$ trivially always the length of the string. So, ignore this.

$Z_2 = 1$	$Z_7 = 0$
$Z_3 = 0$	$Z_8 = 0$
$Z_4 = 0$	$Z_9 = 2$
$Z_5 = 3$	$Z_{10} = 1$
$Z_6 = 1$	$Z_{11} = 0$

Gusfield's Z-algorithm – Defining Z_i-box

Definition of Z-box:

For a string **str**$[1..n]$, **and for any** $i > 1$ **such that** $Z_i > 0$, a Z_i-box is defined as the interval $[i \ldots i{+}Z_i\text{-}1]$ of **str**.



Example

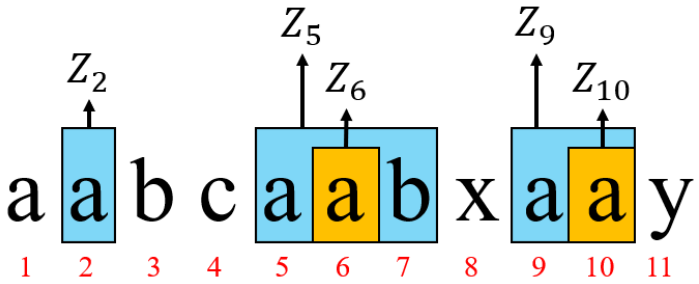
	1	2	3	4	5	6	7	8	9	10	11
str=	a	a	b	c	a	a	b	x	a	a	y

$Z_2 = 1$	$\Longrightarrow$	Z_2-box $= [2..2]$	$Z_7 = 0$	$\Longrightarrow$	Z_7-box = undefined
$Z_3 = 0$	$\Longrightarrow$	Z_3-box = undefined	$Z_8 = 0$	$\Longrightarrow$	Z_8-box = undefined
$Z_4 = 0$	$\Longrightarrow$	Z_4-box = undefined	$Z_9 = 2$	$\Longrightarrow$	Z_9-box $= [9..10]$
$Z_5 = 3$	$\Longrightarrow$	Z_5-box $= [5..7]$	$Z_{10} = 1$	$\Longrightarrow$	Z_{10}-box $= [10..10]$
$Z_6 = 1$	$\Longrightarrow$	Z_6-box $= [6..6]$	$Z_{11} = 0$	$\Longrightarrow$	Z_{11}-box = undefined

Gusfield's Z-algorithm – Defining r_i

Definition of r_i:

For a string **str**$[1..n]$, and for all $i > 1$, r_i is the **right-most endpoint** of all Z-boxes that begin at or before position i.
Alternately, r_i is the largest value of $j+Z_j$-1 over all $1 < j \leq i$, such that $Z_j > 0$.



Example

	1	2	3	4	5	6	7	8	9	10	11
str=	a	a	b	c	a	a	b	x	a	a	y

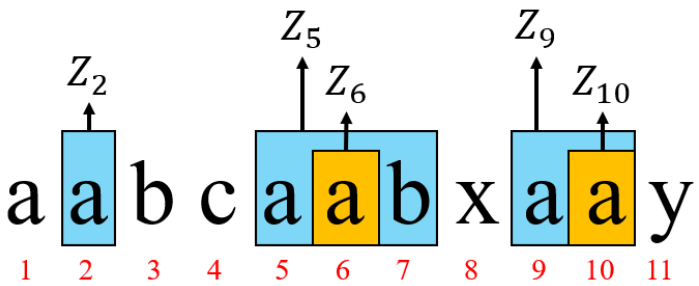
$Z_2 = 1$	$Z_7 = 0$	Z_2-box = [2..2]	$r_2 = 2$	$r_7 = 7$
$Z_3 = 0$	$Z_8 = 0$	Z_5-box = [5..7]	$r_3 = 2$	$r_8 = 7$
$Z_4 = 0$	$Z_9 = 2$	Z_6-box = [6..6]	$r_4 = 2$	$r_9 = 10$
$Z_5 = 3$	$Z_{10} = 1$	Z_9-box = [9..10]	$r_5 = 7$	$r_{10} = 10$
$Z_6 = 1$	$Z_{11} = 0$	Z_{10}-box = [10..10]	$r_6 = 7$	$r_{11} = 10$

Gusfield's Z-algorithm – Defining l_i

Definition of l_i:

For a string **str**$[1..n]$, and for all $i > 1$, l_i is the **left end** of the Z-box that ends at r_i.

In case there is more than one Z-box ending at r_i, then l_i can be chosen to be the left end of **any** of those Z-boxes.

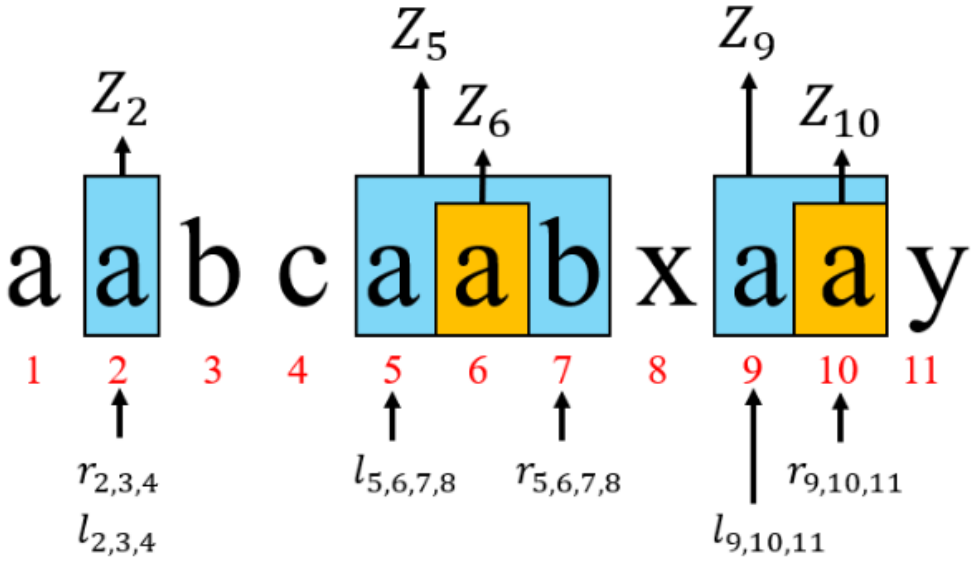


Example

	1	2	3	4	5	6	7	8	9	10	11
str=	a	a	b	c	a	a	b	x	a	a	y

$Z_2 = 1$	$Z_7 = 0$	Z_2-box = [2..2]	$l_2 = 2$	$l_7 = 5$
$Z_3 = 0$	$Z_8 = 0$	Z_5-box = [5..7]	$l_3 = 2$	$l_8 = 5$
$Z_4 = 0$	$Z_9 = 2$	Z_6-box = [6..6]	$l_4 = 2$	$l_9 = 9$
$Z_5 = 3$	$Z_{10} = 1$	Z_9-box = [9..10]	$l_5 = 5$	$l_{10} = 9$
$Z_6 = 1$	$Z_{11} = 0$	Z_{10}-box = [10..10]	$l_6 = 5$	$l_{11} = 9$

Gusfield's Z-algorithm – Example (all things in one slide)



Example

$Z_2 = 1$	Z_2-box $= [2..2]$	$r_2 = 2$	$r_7 = 7$	$l_2 = 2$	$l_7 = 5$
$Z_5 = 3$	Z_5-box $= [5..7]$	$r_3 = 2$	$r_8 = 7$	$l_3 = 2$	$l_8 = 5$
$Z_6 = 1$	Z_6-box $= [6..6]$	$r_4 = 2$	$r_9 = 10$	$l_4 = 2$	$l_9 = 9$
$Z_9 = 2$	Z_9-box $= [9..10]$	$r_5 = 7$	$r_{10} = 10$	$l_5 = 5$	$l_{10} = 9$
$Z_{10} = 1$	Z_{10}-box $= [10..10]$	$r_6 = 7$	$r_{11} = 10$	$l_6 = 5$	$l_{11} = 9$

Main point of Gusfield's Z-algorithm!

- In the previous slides, for any given string, we have defined: $\{Z_i$, Z_i-box, l_i, $r_i\}$.
- Naïve computation of these values, given some string $\texttt{str}[1..n]$ would require $O(n^2)$-time.
- Gusfield's Z-algorithm allows computation of these values, in $O(n)$-time.

Overview of Gusfiled's Z-algorithm

- We compute $Z_i$values **for each successive position** $i \geq 2$ (for 1-based indexing, used in the slides).
- All successively computed array of Z_i values are remembered.
  - Note: Each Z_i-box interval can be computed from its corresponding Z_i value in $O(1)$ time. No need to store the interval information.
- At each iteration, to compute (l_i,r_i) values, you will only needs values of (l_{i-1}, r_{i-1}).
  - Note: no earlier (l_j, r_j) values for $j < i - 1$ are needed ...
  - ...so, temporary variables (l, r) can be used to keep track of the most recently computed (l_{i-1}, r_{i-1}) values to update (l_i,r_i).

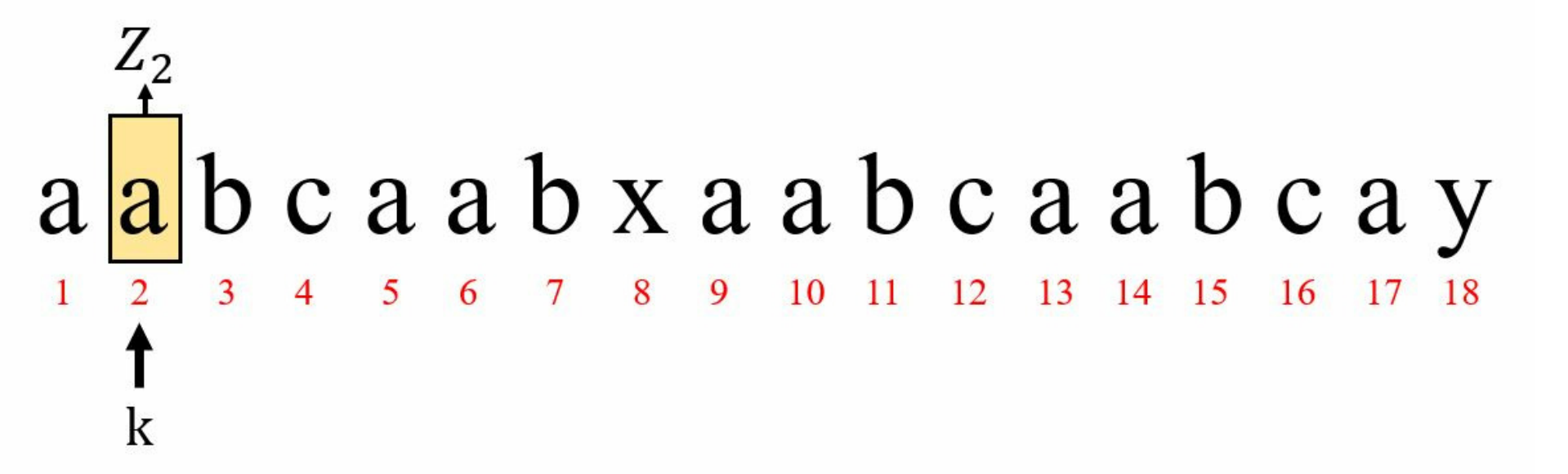
Let's see how this all works out in practice.

(Remember the slides employ 1-based array indexing. In your program, you will have to deal with this in 0-based indexing.)

Z-algorithm – base case

- To begin, compute Z_2 by explicit **left-to-right** comparison of characters **str**$[2 \ldots]$ with **str**$[1 \ldots]$ until a mismatch is found.
- If $Z_2 > 0$
  - set r (i.e., r_2) to $2 + Z_2 - 1 = Z_2 + 1$
  - set l (i.e., l_2) to 2
- else (i.e., if $Z_2 == 0$)
  - set r (i.e., r_2) to 0
  - set l (i.e., l_2) to 0

Z-algorithm – base case example (note: using a new example string from here on)



$Z_2 = 1$ so set $l = 2$ and $r = Z_2 + 1 = 2$.

Z-algorithm – inductive assumption and general cases

Assume inductively . . .

- . . . we have correctly computed the values Z_2 through to Z_{k-1}.
- . . . that r currently holds r_{k-1},
- . . . that l currently holds l_{k-1}.

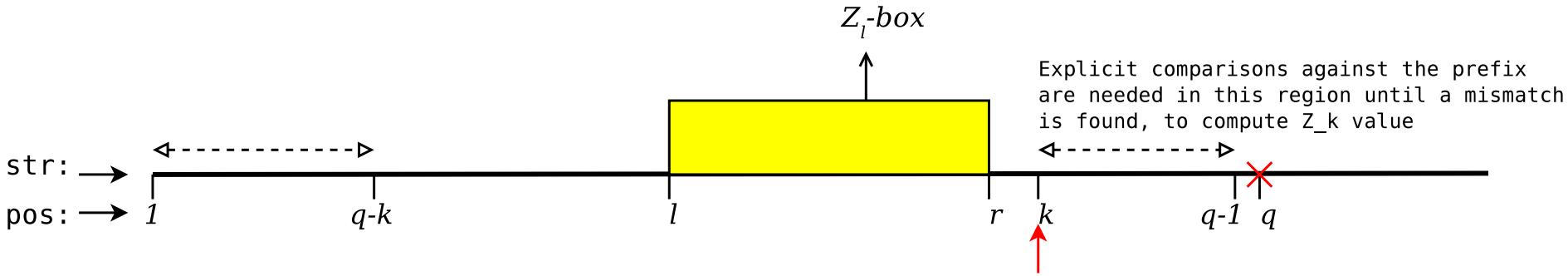
For computing Z_k at position k, these two scenarios arise:

Case 1: If $k > r$

Case 2: Else $k \leq r$

Let's see how to have to handle these two cases.

preprocessing in practice – Case 1: $k > r$



- CASE 1, if $k > r$:
  - Compute Z_k by explicitly comparing characters **str**$[k \ldots q-1]$ with **str**$[1 \ldots q-k]$ until mismatch is found at some $q \geq k$.
  - If $Z_k > 0$:
    - ★ set r (i.e., r_k) to $q-1$.
    - ★ set l (i.e., l_k) to k.
  - Otherwise (i.e. $Z_k = 0$) they retain the same (l, r) values as before

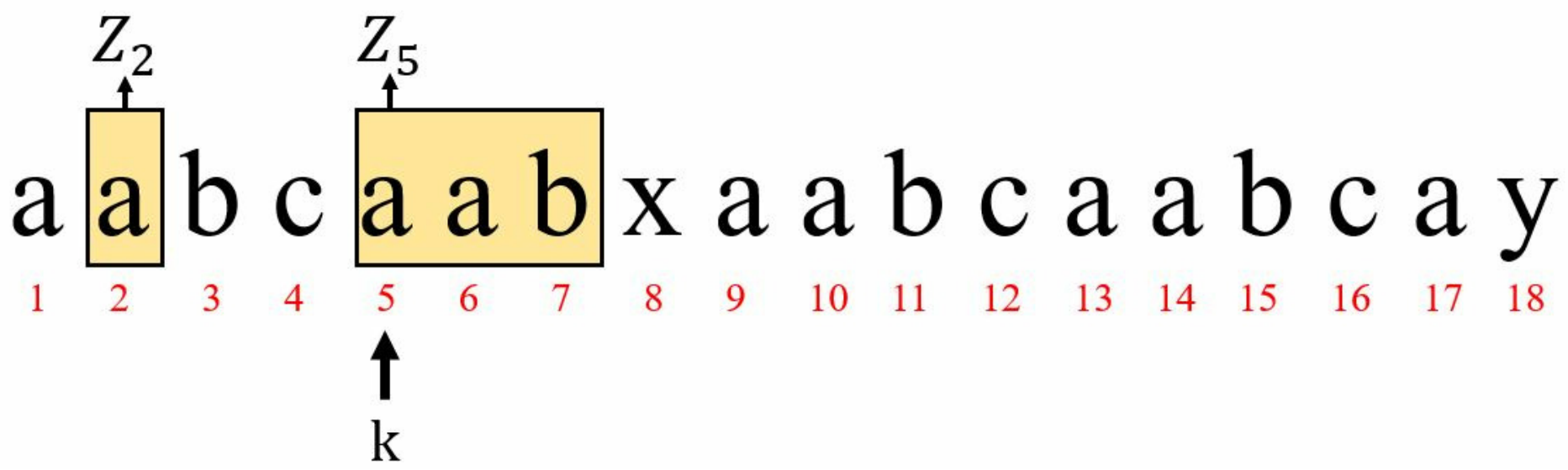
Z-algorithm – Case 1: $k > r$

$Z_3 = 0$ so l and r remain unchanged ($l = 2$ and $r = 2$).

Z-algorithm – Case 1: $k > r$

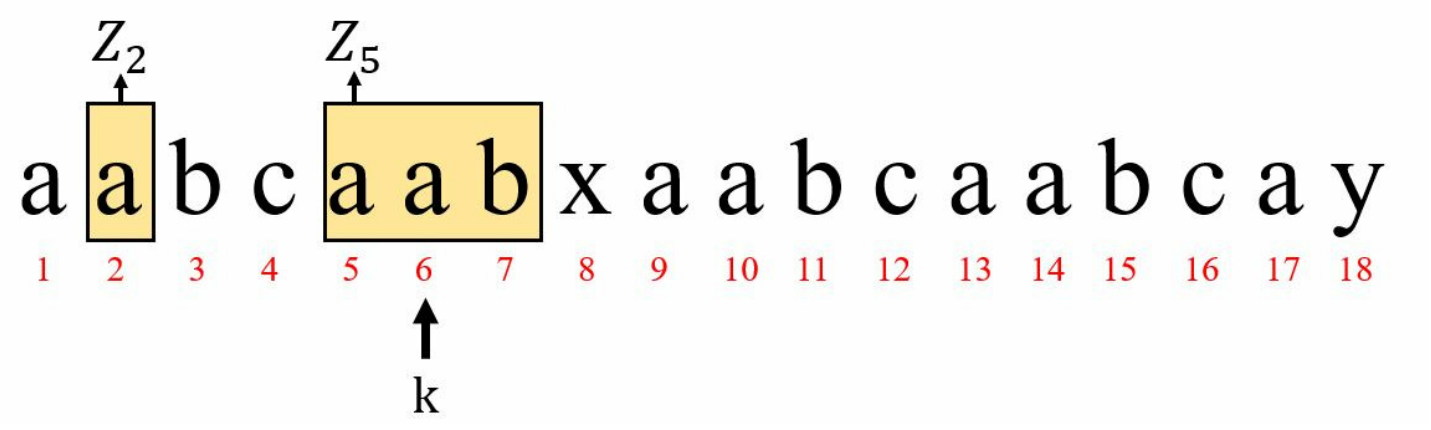
$Z_4 = 0$ so l and r remain unchanged ($l = 2$ and $r = 2$).

Z-algorithm – Case 1: $k > r$



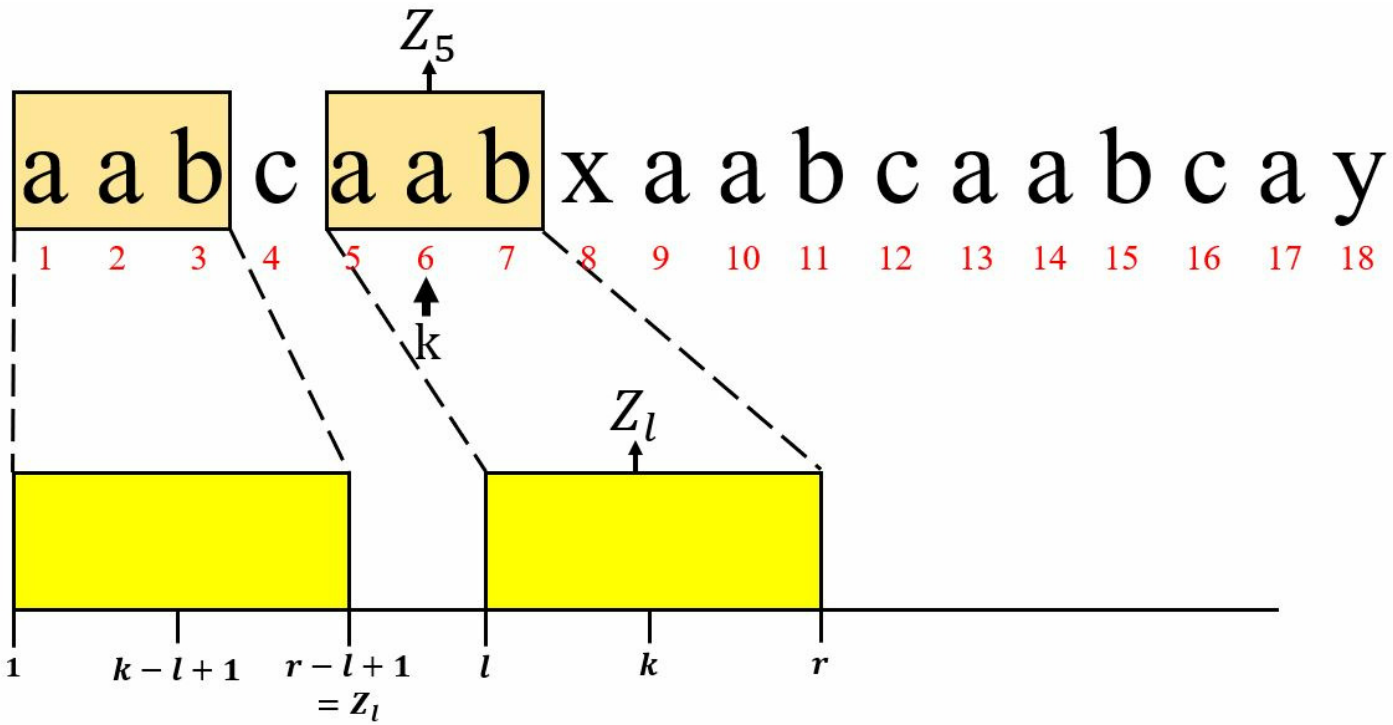
$Z_5 = 3$ so l and r are updated to: $l = 5$ and $r = 7$.

Z-algorithm – Case 2: $k \leq r$



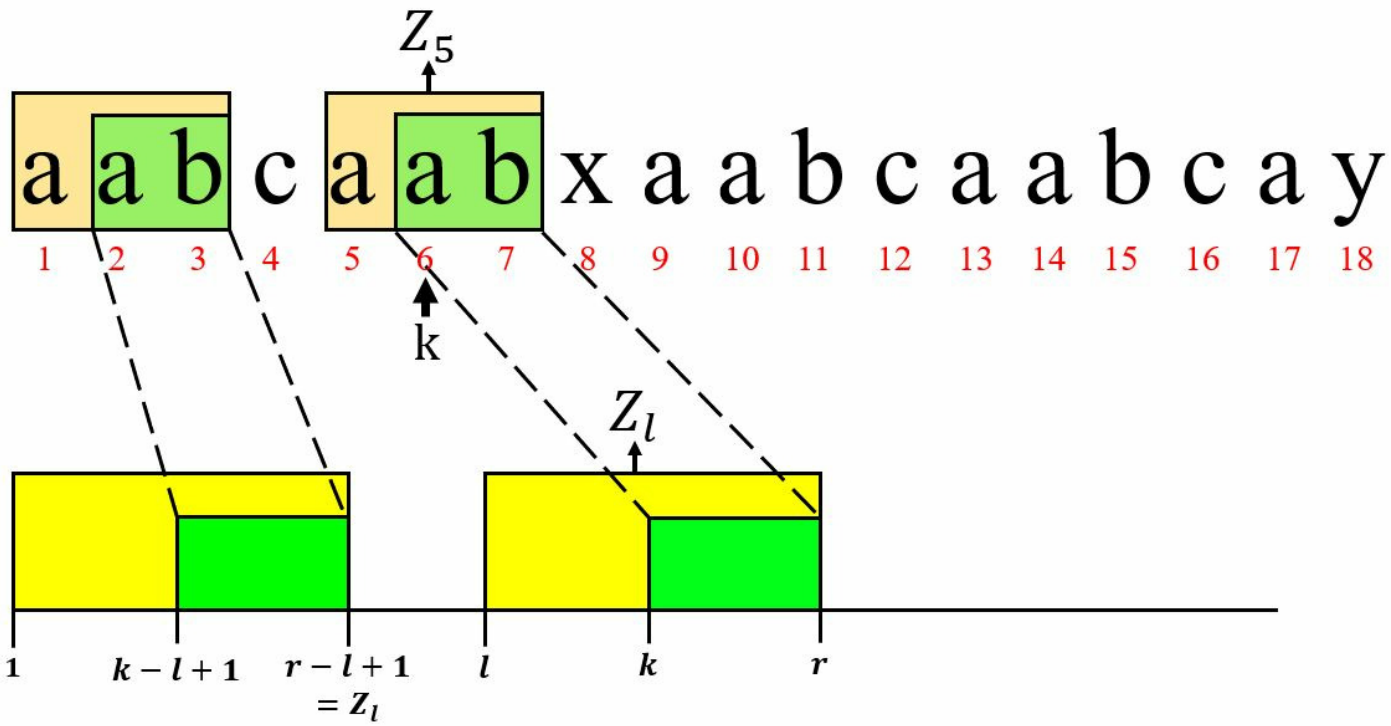
- CASE 2, if $k \leq r$:
  - The character **str**$[k]$ lies anywhere within the substring **str**$[l \ldots r]$ (i.e., anywhere within the Z_l-box).

Z-algorithm – Case 2: $k \leq r$



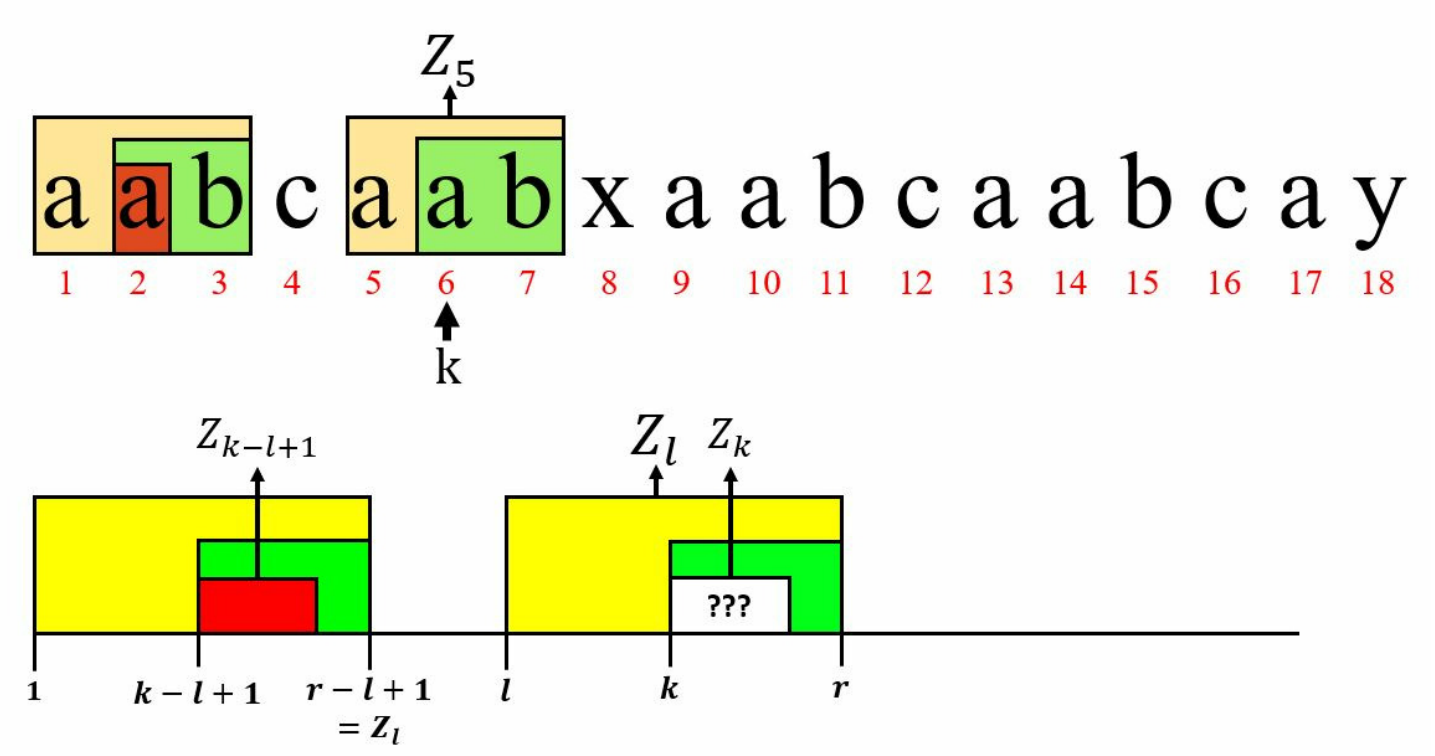
- CASE 2, if $k \leq r$:
  - The character **str**$[k]$ lies anywhere within the substring **str**$[l \ldots r]$ (i.e., anywhere within the Z_l-box).
  - By the definition of Z_l-box, **str**$[l \ldots r]$=**str**$[1 \ldots Z_l]$.

Z-algorithm – Case 2: $k \leq r$



- CASE 2, if $k \leq r$:
  - The character **str**$[k]$ lies anywhere within the substring **str**$[l \ldots r]$ (i.e., anywhere within the Z_l-box).
  - By the definition of Z_l-box, **str**$[l \ldots r]$=**str**$[1 \ldots Z_l]$.
  - This implies the character **str**$[k]$ is identical to **str**$[k-l+1]$
  - By extending this logic, it also implies that the substring **str**$[k \ldots r]$ is identical to **str**$[k-l+1 \ldots Z_l]$.

Z-algorithm – Case 2: $k \leq r$



Question: Can the previously computed value of Z_{k-l+1} (see red box) inform the computation of Z_k (see white box)?

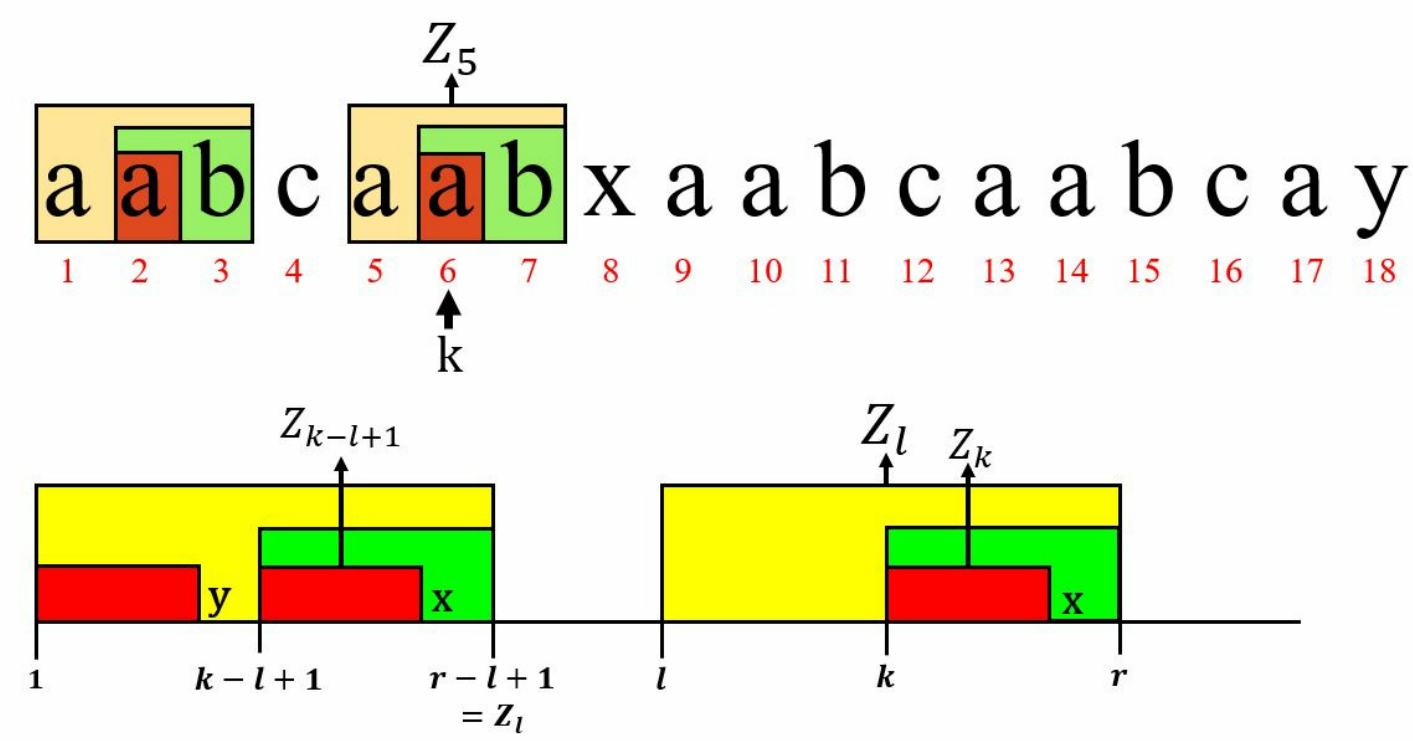
Z-algorithm – overview of handling case 2

In the previous slide, we asked "*can the value of Z_{k-l+1} inform the computation of Z_k?*". The answer is **yes**, and this can be handled by two **sub**-cases: CASES 2a and 2b.

- CASE 2a, if $Z_{k-l+1} < r - k + 1$:
- CASE 2b, else (if) $Z_{k-l+1} \geq r - k + 1$:

Z-algorithm – case 2a mechanics

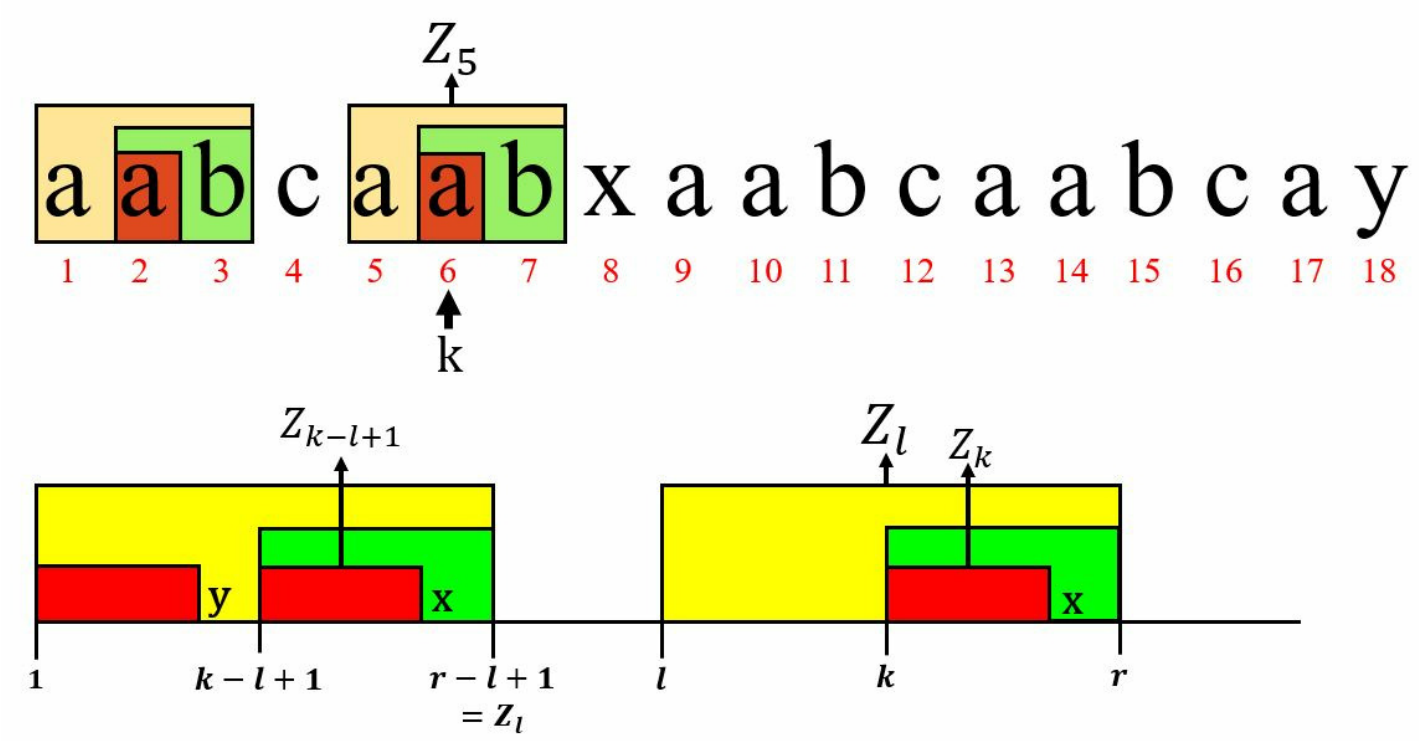
- CASE 2a, if $Z_{k-l+1} < r - k + 1$:



- Therefore, when $Z_{k-l+1} < r - k + 1$
  - set Z_k to Z_{k-l+1}, and
  - r and l values remain unchanged.

Z-algorithm – case 2a mechanics

- CASE 2a, if $Z_{k-l+1} < r - k + 1$:

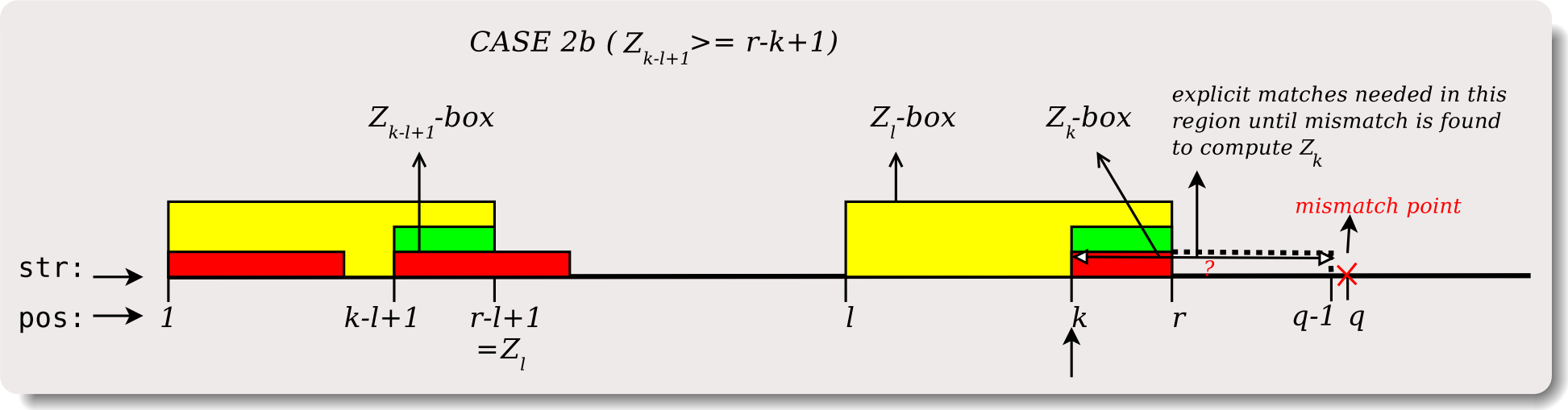


- Therefore, when $Z_{k-l+1} < r - k + 1$
  - set Z_k to Z_{k-l+1}, and
  - r and l values remain unchanged.

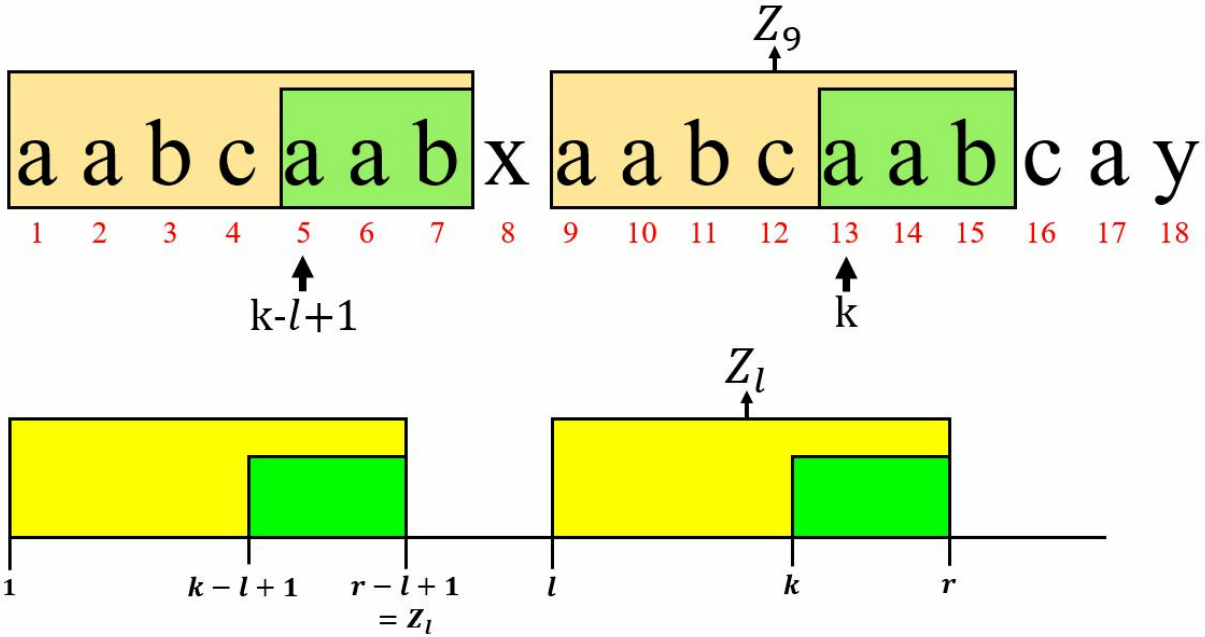
CASE 2a is now complete. Let's now focus on the remaining case 2b.

Z-algorithm – Case 2b

- CASE 2b, if $Z_{k-l+1} \geq r - k + 1$:
  - Z_k must also be $\geq r - k + 1$
  - So, start explicitly comparing **str**$[r+1]$ with **str**$[r-k+2]$ and so on until mismatch occurs.
  - Say the mismatch occurred at position $q \geq r + 1$, then:
    - ★ set Z_k to $q - k$,
    - ★ set r to $q - 1$.
    - ★ set l to k.



Z-algorithm – Case 2b mechanics



- By the definition of Z_l-box, **str**$[l \ldots r]$=**str**$[1 \ldots Z_l]$.
- This implies the character **str**$[k]$ is identical to **str**$[k-l+1]$
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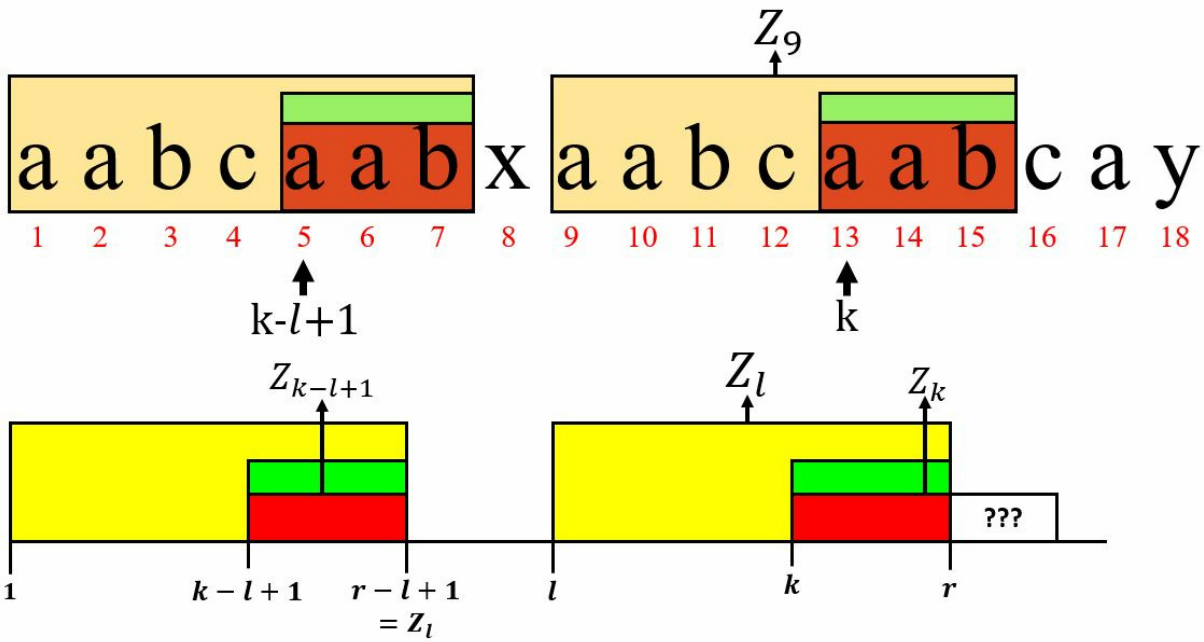

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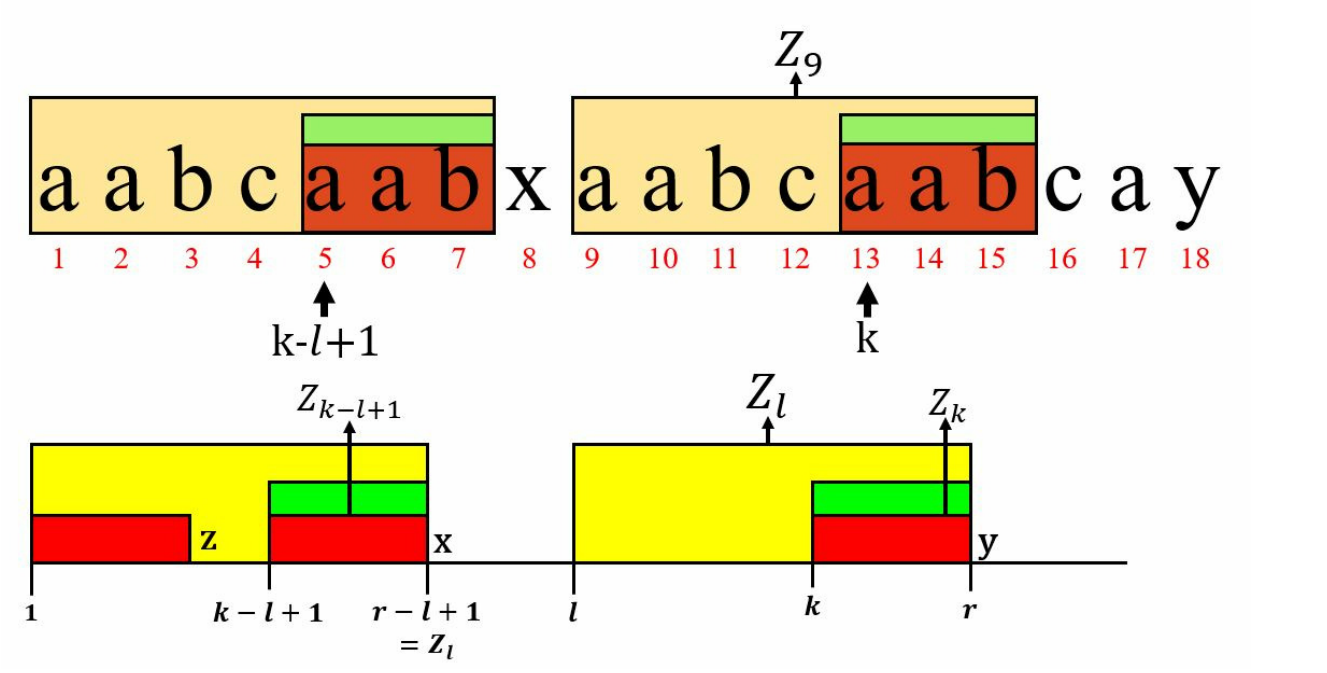

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- What info have we gained so far? Ans: $x \neq y$, $z \neq x$. But we remain uncertain about the relationship between y and z.

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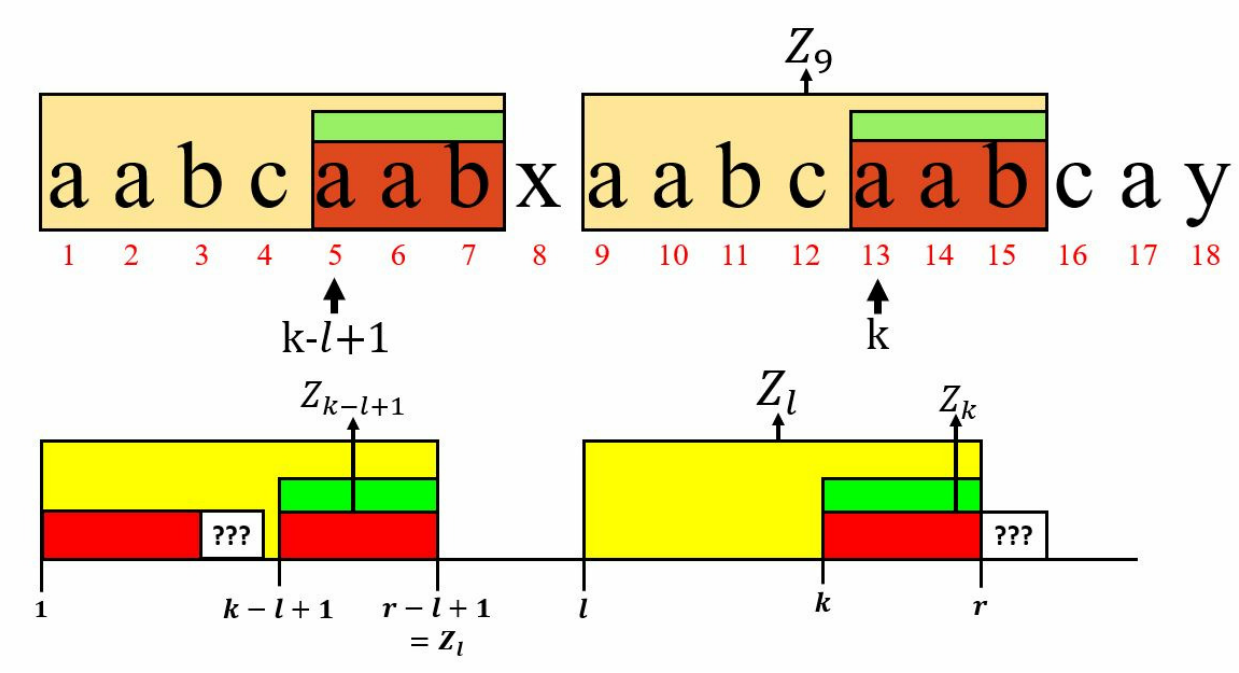

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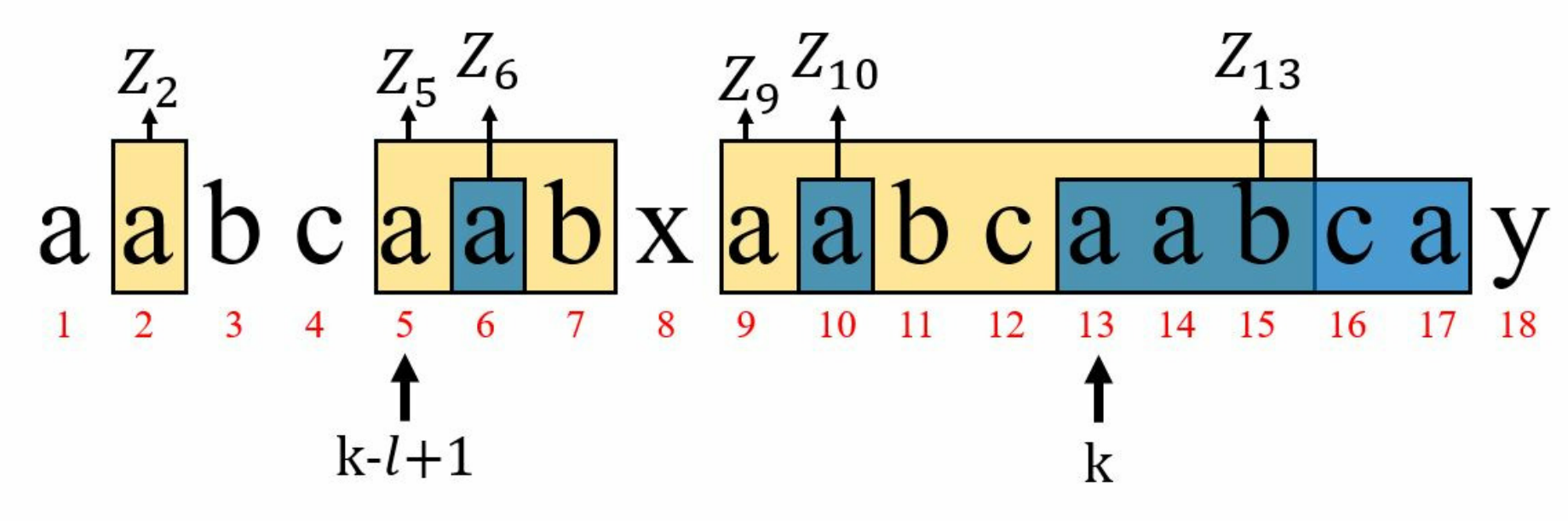
- What info have we gained so far? Ans: $x \neq y$, $z \neq x$. But we remain uncertain about the relationship between y and z.
- Thus, explicit comprisons needed only in the regions of uncertainty.

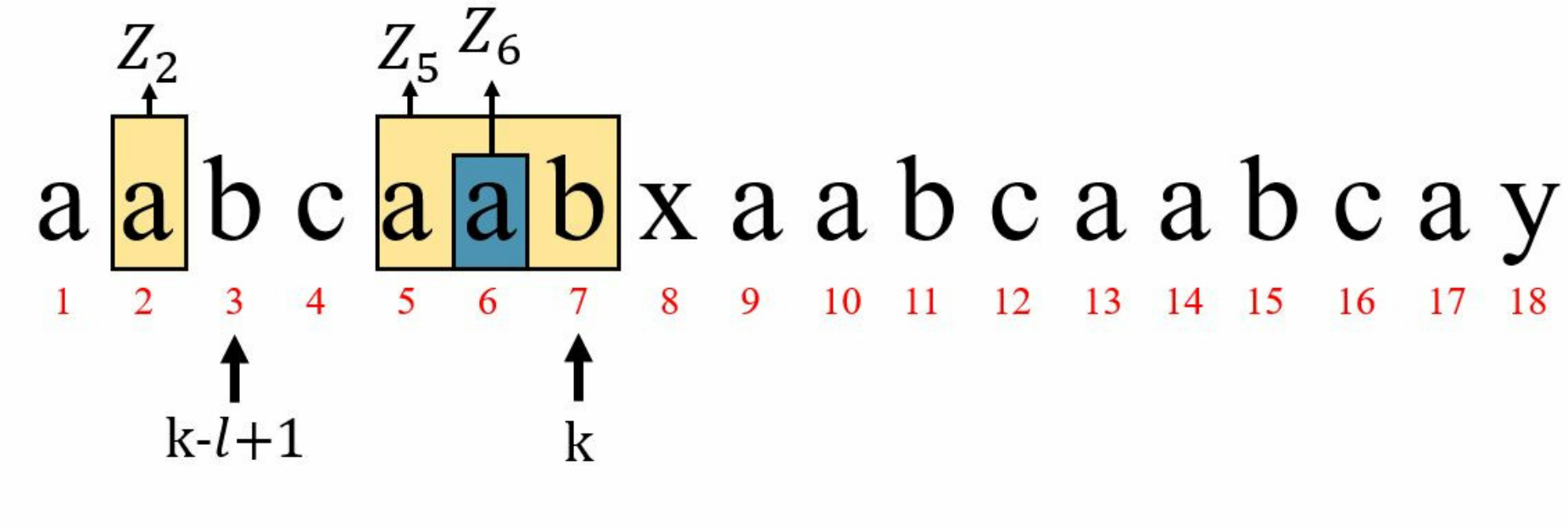
Z-algorithm – Case 2b Example



Perform explicit comparisons until mismatch between characters **str**$[16\ldots]$ and **str**$[4\ldots]$. Initially, $l = 9$ and $r = 15$, so $r + 1 = 16$ and $r - k + 2 = 4$. After computation of Z_{13}, the l and r values are updated to $l = 13$ and $r = 17$.

Z-algorithm: continuing iterations beyond those already covered – self-study

On the slide series #26 and #29, examples iterations $k = 2 \ldots 5$ and $k = 6$ were covered, respectively. Below we have continue the running example for $k \geq 7$.



Case 2a: $Z_7 = 0$ so l and r remain unchanged ($l = 5$ and $r = 7$).

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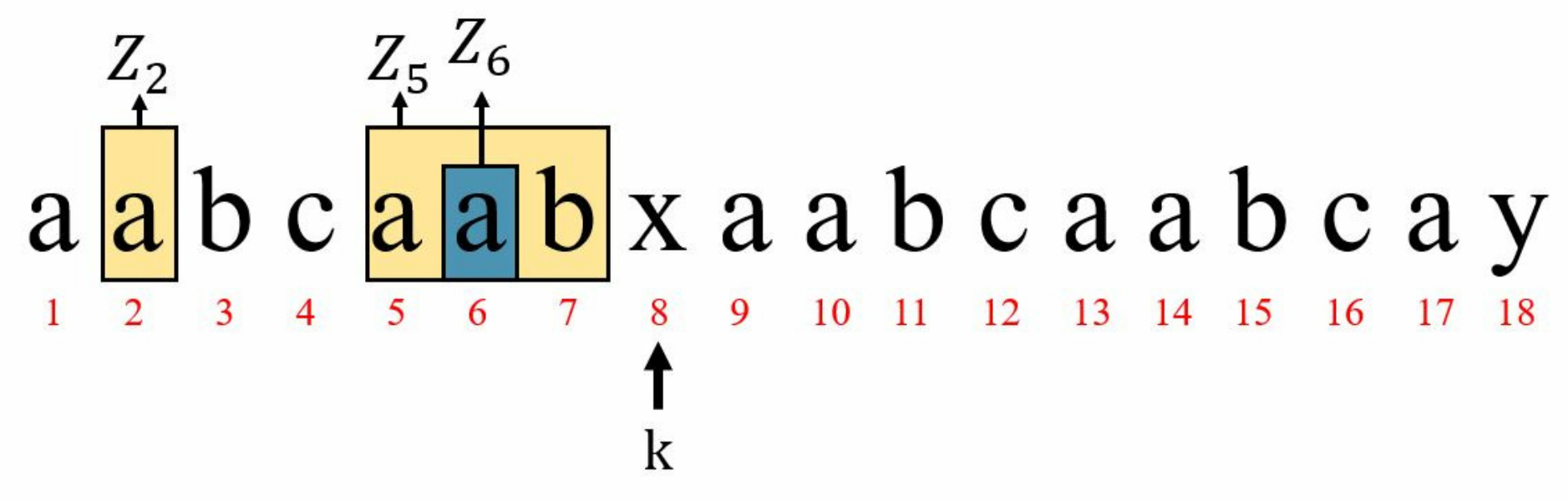

Case 1: $Z_8 = 0$ so l and r remain unchanged ($l = 5$ and $r = 7$).

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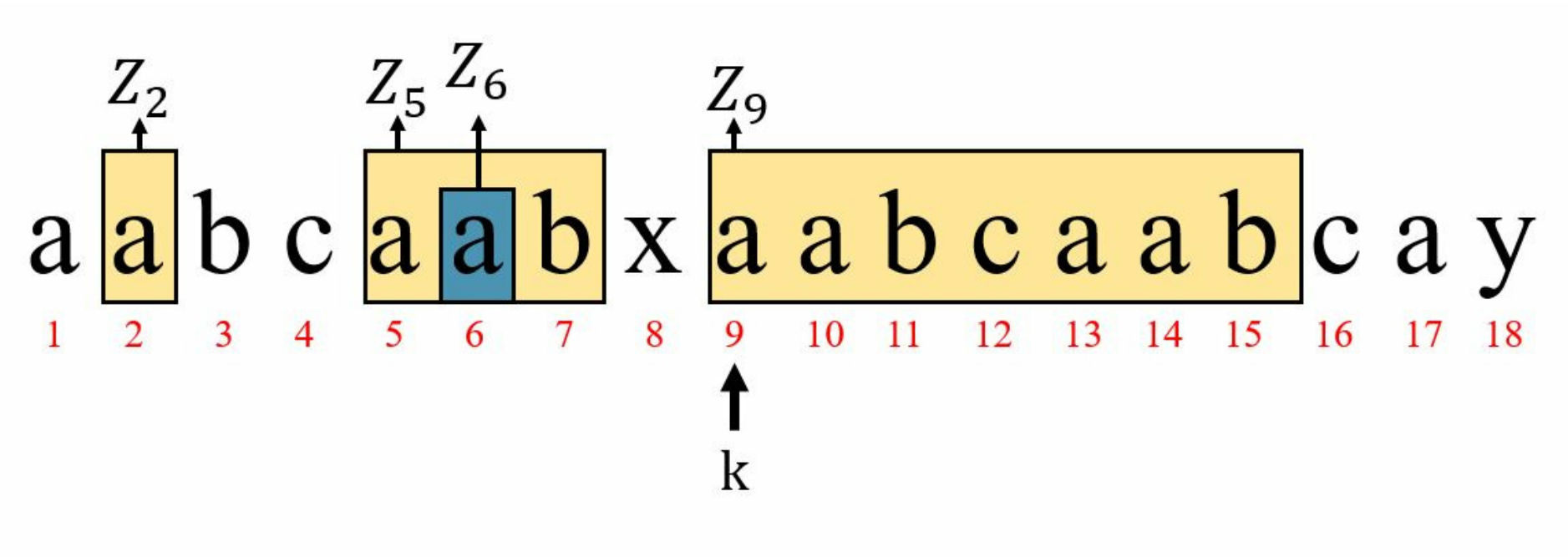

Case 1: $Z_9 = 7$ so l and r are updated to: $l = 9$ and $r = 15$.

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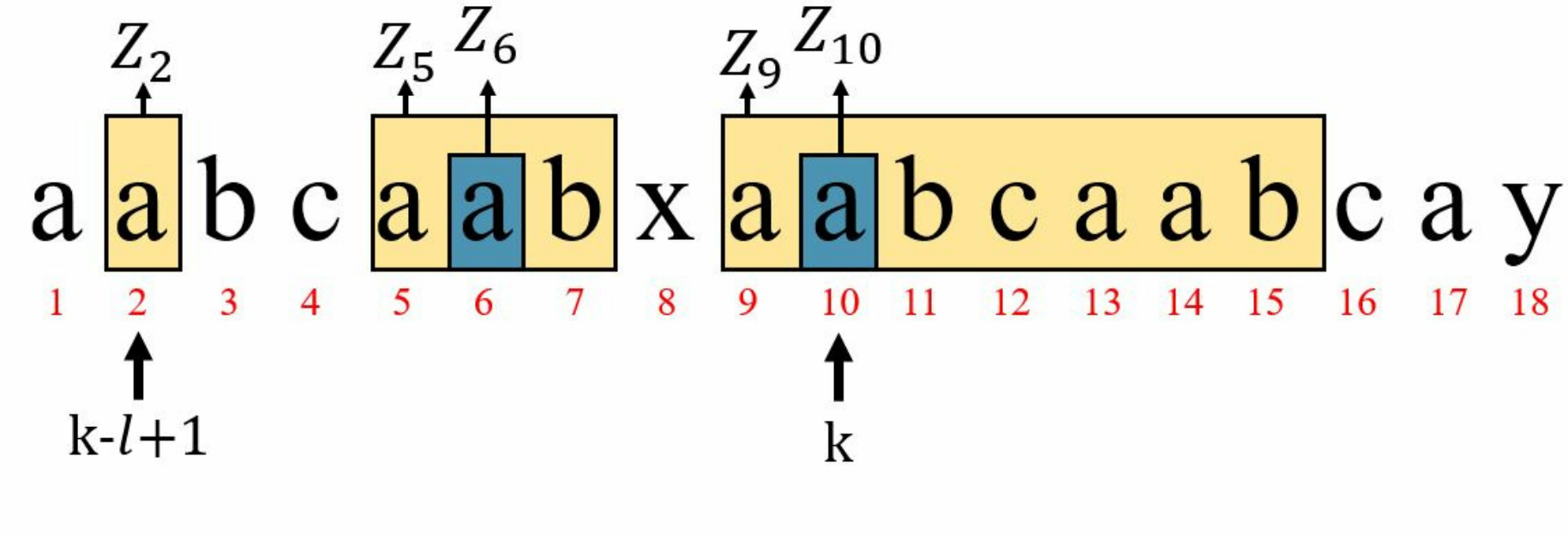

Case 2a: $Z_{10} = 1$ so l and r remain unchanged ($l = 9$ and $r = 15$).

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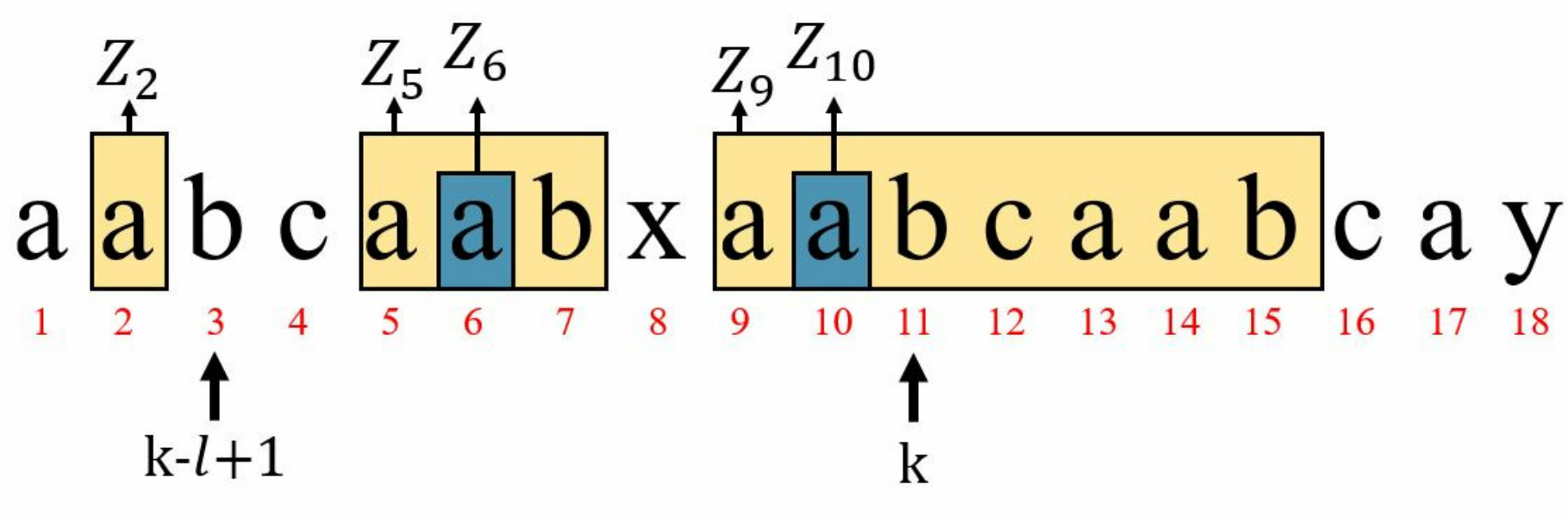

Case 2a: $Z_{11} = 0$ so l and r remain unchanged ($l = 9$ and $r = 15$).

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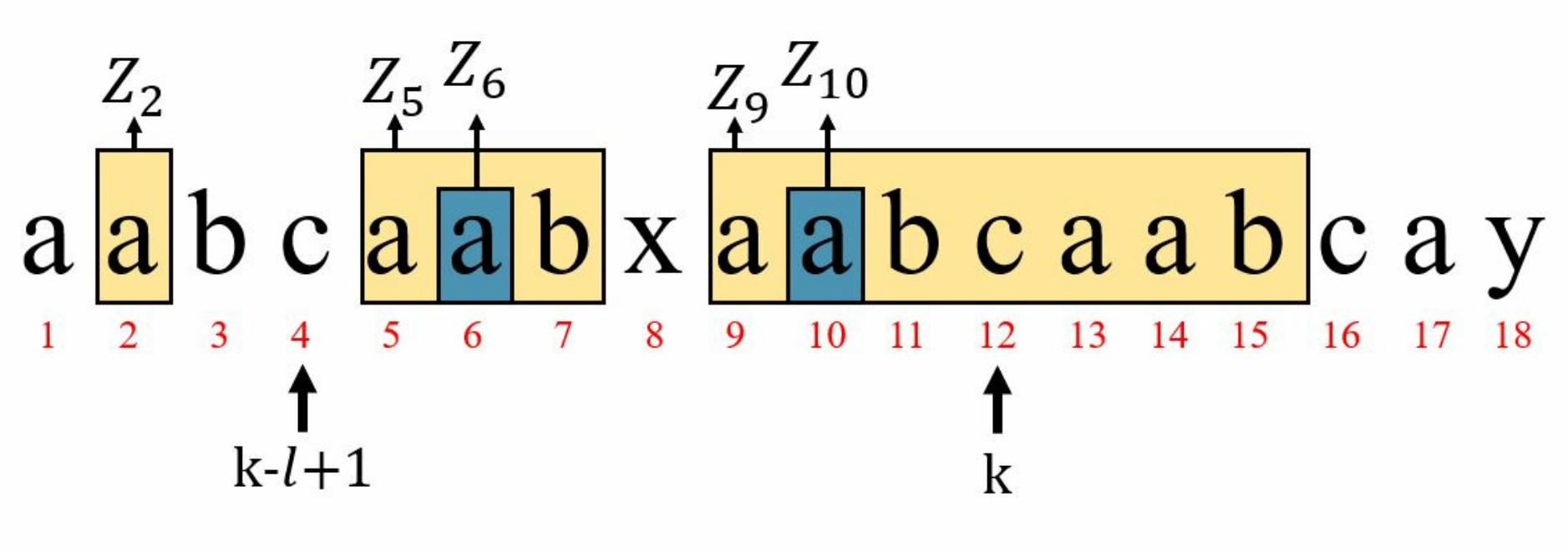

Case 2a: $Z_{12} = 0$ so l and r remain unchanged ($l = 9$ and $r = 15$).

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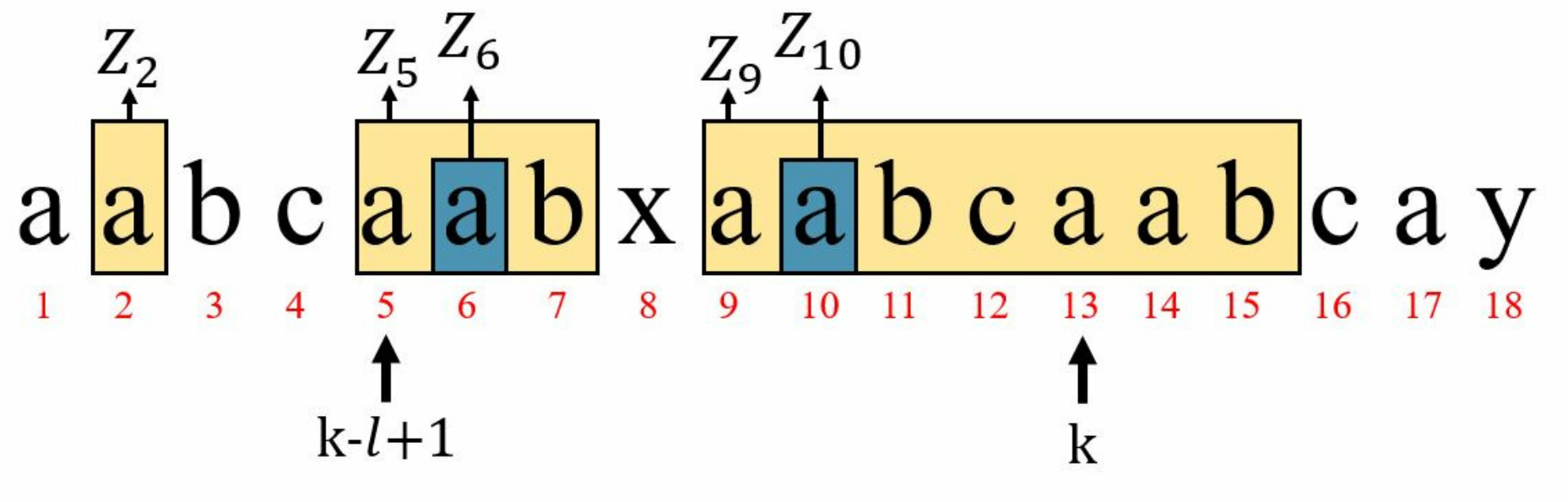

Case 2b (see slide #32): $Z_{13} = 5$ (but needs only 3 extra comparisons, half of those required for the naive approach).

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  - But $r_k \leq n$.
  - Thus, there are at most n **matches**.

Using Z-algorithm for linear-time exact pattern matching

Recall the exact pattern matching problem

Given a reference text **txt**$[1 \ldots n]$ and a pattern **pat**$[1 \ldots m]$, find **ALL** occurrences, if any, of **pat** in **txt**.

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Realizing a linear-time solution using Gusfield's Z-algorithm/preprocessing

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- We already, showed that the computation of Z_i values for any sting **str** takes $O(|$**str**$|)$ time.
- Thus, this pattern matching algorithm takes $O(m + n)$ time. **QED**

Lecture Summary

- We learnt Gusfield's Z-algorithm for string preprocessing.
- It has versatile uses, including pattern matching.
- Naïve pattern matching takes $O(m \times n)$-time.
- Z algorithm guarantees pattern matching in $O(n + m)$-time.

Next week ...

we will learn and analyze

- the famous Boyer-Moore's algorithm for pattern matching
  - which takes $O(n + m)$-time worst case ...
  - ... but $O(\frac{n}{m})$-time in most 'realworld' usage.
- the Knuth-Morris-Pratt algorithm also takes $O(n + m)$-time worst-case, but inferior in performance to Boyer-Moore in practice.

For both, we will make use of the Z-algorithm we learnt here. So, as pre-class learning for next week, do not forget to fully internalize the Z-algorithm.

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END

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